

Project Report: End-to-End Business Metrics Platform - Data Analyst

Written by Gagan Pasupuleti

Family:	Data Analyst
Level:	Advanced
Chapters:	6
Source:	CodeQuest project specification (official docs + industry patterns)

Summary

Automated weekly KPI report: SQL extraction, Python pipeline, validation checks, and email delivery.

Chapters

Track: Project-Intro

Chapter 1: Project Overview [Advanced]

Chapter focus: Project Overview** *(Level: Advanced)

End-to-End Business Metrics Platform** is a advanced-level end-to-end project for Data Analyst. **Duration:** 4-5 weeks. **Automated weekly KPI report:** SQL extraction, Python pipeline, validation checks, and email delivery. **Tools:** Python, SQL, Great Expectations, Jinja2.

Code Reference:

```
Objective: Automate data extraction with Python
Objective: Data quality validation rules
Objective: Scheduled report generation
Objective: Document data dictionary
```

What it shows: Lists measurable outcomes before you start building.

Topic guide

What it actually is

A project overview defines scope, tools, and success criteria.

When to use it

At project kickoff.

Where to use it

Data Analyst portfolios and CodeQuest project submissions.

Real use example

A candidate lists this project on a Data Analyst resume with a demo link.

Track: Project-Phase

Chapter 2: Phase: Pipeline [Advanced]

Chapter focus: Phase: Pipeline** *(Level: Advanced)

Extract-transform-load script. Complete these tasks: SQL ? pandas; Schedule cron. Deliverable for this phase: pipeline.py.

Code Reference:

- [] SQL ? pandas
- [] Schedule cron

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Pipeline is one step in the full project lifecycle.

When to use it

During week 1 of the project timeline.

Where to use it

Data Analyst agile sprints and coursework milestones.

Real use example

Team demo shows the pipeline.py to the teacher.

Chapter 3: Phase: Quality [Advanced]

Chapter focus: Phase: Quality** *(Level: Advanced)

Great Expectations checks. Complete these tasks: Null rate; Row count. Deliverable for this phase: Quality report.

Code Reference:

- [] Null rate
- [] Row count

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Quality is one step in the full project lifecycle.

When to use it

During week 2 of the project timeline.

Where to use it

Data Analyst agile sprints and coursework milestones.

Real use example

Team demo shows the Quality report to the teacher.

Chapter 4: Phase: Report [Advanced]

Chapter focus: Phase: Report *(Level: Advanced)**

Auto-generate PDF/HTML. Complete these tasks: Jinja template; Charts embedded. Deliverable for this phase: Sample weekly report.

Code Reference:

- [] Jinja template
- [] Charts embedded

What it shows: Checklist format helps track phase completion.

Topic guide**What it actually is**

Phase Report is one step in the full project lifecycle.

When to use it

During week 3 of the project timeline.

Where to use it

Data Analyst agile sprints and coursework milestones.

Real use example

Team demo shows the Sample weekly report to the teacher.

Chapter 5: Phase: Docs [Advanced]

Chapter focus: Phase: Docs *(Level: Advanced)**

Data dictionary and lineage. Complete these tasks: Column definitions; Source mapping. Deliverable for this phase: Data catalog doc.

Code Reference:

- [] Column definitions
- [] Source mapping

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Docs is one step in the full project lifecycle.

When to use it

During week 4 of the project timeline.

Where to use it

Data Analyst agile sprints and coursework milestones.

Real use example

Team demo shows the Data catalog doc to the teacher.

Track: Project-Deliverables

Chapter 6: Final Deliverables and Review [Advanced]

Chapter focus: Final Deliverables and Review** *(Level: Advanced)

Submit all final deliverables: Pipeline code; Quality report; Sample weekly report; Data dictionary. Skills demonstrated: ETL, Data quality, Automation. Prepare a 5-minute walkthrough video and README for CodeQuest teacher marking.

Code Reference:

```
Deliverable: Pipeline code
Deliverable: Quality report
Deliverable: Sample weekly report
Deliverable: Data dictionary
```

What it shows: Final checklist ensures nothing is missing at submission.

Topic guide

What it actually is

Deliverable review confirms end-to-end project completion.

When to use it

Before deadline and teacher review.

Where to use it

CodeQuest project gallery and interviews.

Real use example

A student submits Pipeline code and passes the rubric.